Self-Organized Criticality in Social Media Information Cascades: A Network-Based Complexity Study

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Abstract

Self-organized criticality (SOC) is a fundamental property of complex systems characterized by scale-invariant behavior and power-law distributions. In this work, we investigate SOC in social media systems by modeling information cascades on a complex network. Using stochastic simulations, we demonstrate that the distribution of cascade sizes follows a power-law, indicating critical dynamics. We explicitly define the roles of system noise, network connectivity, and resulting avalanches. Furthermore, the relationship between varying network densities and the frequency distribution of cascades is analyzed, confirming that social media systems naturally evolve toward a critical state without the need for external parameter tuning.

1 Introduction

The paradigm of complexity science emerged as a necessary departure from classical reductionism, which historically dominated the natural sciences. Reductionism, predicated on the Cartesian paradigm, posits that a complex system can be understood by dissecting it into its constituent components and analyzing them in isolation. While this approach has been profoundly successful in classical mechanics and linear systems, it systematically fails when applied to systems characterized by nonlinear interactions, feedback loops, and emergent phenomena. The roots of complexity science can be traced back to Henri Poincaré's work on the three-body problem in the late nineteenth century, which shattered the deterministic predictability of Newtonian mechanics by introducing the concept of deterministic chaos. This realization lay dormant until the middle of the twentieth century when the advent of computational power allowed pioneers like Edward Lorenz to empirically demonstrate chaos theory through the strange attractors in weather systems. Subsequently, the establishment of the Santa Fe Institute in the 1980s galvanized interdisciplinary efforts to formally study complex adaptive systems, uniting physics, biology, computer science, and economics. Complexity science now provides the mathematical and conceptual scaffolding for understanding systems where the collective behavior of parts is fundamentally different from the sum of those parts.

Within this framework, self-organized criticality (SOC) occupies a cornerstone position. Self-organized criticality, first introduced by Bak, Tang, and Wiesenfeld (1987), describes non-equilibrium systems that naturally evolve to a critical state. Classic examples include sandpile models, earthquakes, and forest fires. In recent years, digital ecosystems—particularly social media platforms—have exhibited similar macroscopic

behaviors. Small, localized events (such as a single user posting an image) occasionally trigger massive, global cascades (virality). This study applies complexity science to model these information cascades, demonstrating that they are not mere anomalies, but rather the expected avalanches of a system in a self-organized critical state. By systematically examining these phenomena, this work aims to bridge theoretical physics and computational social science.

2 Literature Review

The formalization of complexity in digital networks requires grounding in three interconnected domains: the theoretical foundations of self-organized criticality, the historical evolution of network graph theory, and contemporary empirical studies on digital information propagation and virality.

2.1 Self-Organized Criticality and the BTW Sandpile Model

The concept of self-organized criticality fundamentally revolutionized statistical mechanics by offering an endogenous mechanism for the generation of fractal geometries and $1/f$ noise in nature. Prior to SOC, it was assumed that extended critical states (displaying scale invariance) could only be achieved if an external observer carefully tuned control parameters, such as bringing a thermodynamic system precisely to its critical temperature. The Bak-Tang-Wiesenfeld (BTW) sandpile model upended this assumption. In their seminal 1987 paper, Bak, Tang, and Wiesenfeld demonstrated that certain non-equilibrium open systems, driven slowly by an external force (sand grains dropping probabilistically on a grid), will naturally self-organize into a highly sensitive, critical resting state. Once this state is achieved, a single additional grain can trigger an avalanche of any size, constrained only by the finite dimensions of the system grid.

The probability distribution of the sizes of these avalanches follows a strict power-law. In this regime, standard statistical measures such as mean and variance lose their predictive utility, as the heavy-tailed nature of the distribution implies that catastrophic, system-wide events do not sit outside the normal operating parameters of the system—they are an integral feature of its dynamics. Since its introduction, the BTW paradigm has been utilized to explain a staggering diversity of phenomena, from the Gutenberg-Richter law of earthquake magnitudes to the fluctuations of financial markets and the extinction rates in evolutionary biology. In the context of the present research, the framework of SOC provides the foundational hypothesis that social media networks internally regulate their connectivity to hover near a critical state, perpetually poised for viral avalanches.

2.2 The Evolution of Network Theory

Understanding how localized cascades propagate requires a sophisticated theory of the underlying topological substrate. The formal study of networks originated with Leonhard Euler's resolution of the Seven Bridges of Königsberg in 1736, birthing graph theory. However, the modern paradigm of complex networks began primarily with the Erdős–Rényi (ER) model formulated in the late 1950s. The ER model generated random graphs by connecting nodes with a constant probability, resulting in a Poisson degree distribution. While mathematically elegant, the ER model failed to capture the architectural realities

of natural and social networks, notably lacking both high clustering (the tendency of a node's neighbors to be connected to each other) and the existence of highly connected nodes.

A monumental leap occurred in 1998 when Duncan Watts and Steven Strogatz introduced the "small-world" network model. By taking highly structured lattice rings and rewiring a small fraction of edges randomly, they demonstrated how high local clustering could coexist with remarkably short average path lengths—the mathematical formalization of the "six degrees of separation" phenomenon. Yet, the Watts-Strogatz model still produced homogeneous degree distributions. This limitation was resolved in 1999 by Albert-László Barabási and Réka Albert, who identified that real-world networks—such as the World Wide Web, citation networks, and biological metabolic pathways—systematically exhibit power-law degree distributions. They proposed the Scale-Free network model, driven by two mechanisms: continuous growth and preferential attachment (the "rich-get-richer" phenomenon). The Barabási-Albert (BA) model fundamentally redefined network science, establishing that networks are not static entities but dynamically evolving architectures where extreme heterogeneity in node connectivity structurally dictates system resilience, vulnerability, and contagion dynamics.

2.3 Homophily, Echo Chambers, and Virality on Social Media

With the proliferation of Web 2.0, complexity paradigms were rapidly adapted to study social media architectures. Academic research over the last decade has heavily focused on the intertwined phenomena of information diffusion, homophily, and the formation of echo chambers. Homophily—the sociological principle that "birds of a feather flock together"—drives users to disproportionately connect with ideologically similar agents. When this dynamic is combined with algorithmic recommendation systems designed to maximize user engagement (thereby exploiting cognitive biases), the network frequently partitions into self-reinforcing, polarized clusters known as echo chambers.

Recent efforts to model these dynamics, such as those employing bounded confidence models or multi-agent simulations, have revealed that echo chambers act as both incubators and barriers to information spread. Highly clustered echo chambers foster intense, rapid local cascades but often impede the global diffusion of ideas across the broader network boundaries. Conversely, virality—the rare, global explosion of a piece of information—depends intricately on the bridge nodes or "weak ties" that span these distinct topological clusters, as famously theorized by Mark Granovetter. Contemporary epidemiological models adapted for social media (such as complex contagion models where multiple exposures are required for transmission) demonstrate that the structural inequalities characterized by BA scale-free networks are prerequisites for the extreme volatility observed in digital attention economies. This intersection of network topology, algorithmic curation, and human cognitive biases establishes the rigorous background against which our specific SOC simulations are deployed.

3 Social Media as a Complex System

To apply complexity science, a system must exhibit interactions that produce emergent behavior not predictable by analyzing individual components in isolation. Social media is an ideal candidate for this framework because:

- **Large-scale Interacting Agents:** Millions of users act as nodes, making independent, localized decisions based on limited information. From a cognitive perspective, these agents operate under severe constraints of bounded rationality and finite attention spans. As theorized by behavioral economists and cognitive psychologists, individuals navigating an environment characterized by extreme information overload rely on rapid, often unconscious heuristics. The digital architecture forces these agents to process vast streams of stimuli, collapsing complex ideological or aesthetic evaluations into binary micro-actions (likes, shares, or retweets). This compression of human psychology into discrete, stochastic networking actions forms the micro-level foundation upon which macro-level complexity is built.

- **Nonlinear Amplification:** A post shared by a highly connected node can non-linearly amplify the reach of the information. In a standard Gaussian distribution of influence, individual actions have proportionally equivalent impacts on the system state. However, owing to the extreme scale-free topology of digital networks, amplification processes are highly non-linear. The threshold models of collective behavior illustrate that moving a piece of information from a peripheral user to a heavily connected "hub" node (an influencer) transforms the topological distance of that information to the rest of the network practically to zero. A single interaction by such a hub can instantaneously catalyze an exponential explosion in exposure, rendering traditional linear cause-and-effect reasoning entirely obsolete.

- **Dynamic Topology:** The network is not static; users constantly follow and unfollow others, dynamically rewiring the system based on attention. Theoretical models of adaptive networks suggest that the topology itself and the state of the agents co-evolve across time. Users prune connections that yield dissonant or uninteresting information, while enthusiastically foraging for new connections that optimize their localized attention rewards. This constant, decentralized structural plasticity means the network possesses "memory" and is capable of topological learning. The architecture at any time step is both the constrained product of past interactions and the rigid framework dictating the cascades of the future, creating an intractable, recursive feedback loop.

- **Emergence:** Virality cannot be predicted by looking at a single user; it is an emergent property of the entire network's current state. Taking inspiration from Thomas Schelling's models of macro-behavior arising from disjointed micro-motives, virality fundamentally resists reductionist prediction. One cannot dissect a meme or a post, analyze its semantic content, and accurately forecast its trajectory. Rather, a viral cascade is a phase transition—an emergent outcome dependent on the precise stochastic pathways of early adopters, the specific and transient vulnerabilities of network clusters, and the instantaneous global attention state. The phenomenon of the cascade strictly belongs to the whole system and vanishes entirely when the system is empirically deconstructed.

3.1 Master Equation for Network Evolution

To formalize the dynamic topology underlying this complex system, we turn to the continuum theory rate equation, or macro-state master equation, for the Barabási–Albert scale-free network. If we assume the network grows continuously by adding nodes with

m edges, the rate of change of the degree k_i of node i under preferential attachment is given by:

$$\frac{\partial k_i}{\partial t} = m \frac{k_i}{\sum_j k_j} \tag{1}$$

Since the total number of edges at time t is approximately mt, the sum of all degrees is $\sum_j k_j = 2mt$. Substituting this yields $\frac{\partial k_i}{\partial t} = \frac{k_i}{2t}$. Integrating this differential equation with the initial condition $k_i(t_i) = m$ at the time t_i the node was introduced, we obtain the time evolution of the node's degree: $k_i(t) = m\left(\frac{t}{t_i}\right)^{\frac{1}{2}}$.

To find the asymptotic degree distribution $P(k)$, we formulate the probability that a node has a degree smaller than k: $P(k_i(t) < k) = P\left(t_i > \frac{m^2 t}{k^2}\right)$. Assuming nodes are added at uniformly distributed times, the probability density function for t_i is $\frac{1}{t}$. By evaluating the derivative of the cumulative distribution function, we arrive at the steady-state degree distribution:

$$P(k) = \frac{\partial P(k_i(t) < k)}{\partial k} = \frac{\partial}{\partial k}\left(1 - \frac{m^2}{k^2}\right) = 2m^2 k^{-3} \tag{2}$$

This exact derivation mathematically proves the $P(k) \propto k^{-3}$ scaling, establishing the extreme heterogeneity that acts as the physical substrate for critical information avalanches.

4 System Mechanics: Noise, Connectivity, and Avalanches

To model this system, we map the core tenets of SOC to social media dynamics:

4.1 Connectivity (The Grid)

The system is modeled as a graph $G = (V, E)$, where V represents users and E represents the "follower" connections between them. Unlike a simple lattice, social media connectivity is modeled using a Scale-Free (Barabási–Albert) network, governed by preferential attachment. The probability $\Pi(k_i)$ that a new node connects to an existing node i with degree k_i is given by:

$$\Pi(k_i) = \frac{k_i}{\sum_j k_j} \tag{3}$$

This "rich-get-richer" mechanism is precisely what leads to the highly skewed, heavy-tailed degree distributions characteristic of social media networks. This means a few nodes (influencers) have massive connectivity, while most nodes have very few edges. A visual representation of this topology is shown in Figure 1 (Left).

4.2 Noise (The Perturbation)

In a sandpile model, noise is the slow, random dropping of sand grains. In our system, **noise** is the spontaneous generation of new information. At any given time step, a random user may generate a new post with a small probability. Furthermore, the decision of a neighboring node to share the post is governed by a stochastic transmission probability, p.

4.3 Avalanches (The Cascade)

When a post is created (a grain of sand drops), it may be ignored, or it may be shared by neighbors, who then expose it to their neighbors. An **avalanche** is the resulting information cascade. The size of the avalanche, S, is defined as the total number of unique nodes that share the information before the cascade dies out.

$$S = \sum_{t=0}^{T} N_t \tag{4}$$

where N_t is the number of newly activated nodes at time step t, and T is the time when the cascade terminates.

4.3.1 Mathematical Formulation as a Branching Process

The propagation of the information avalanche can be rigorously defined using a branching process framework. Let p_k be the probability distribution of secondary infections produced by a single node. We define the primary generating function for the degree distribution as $G_0(x) = \sum_{k=0}^{\infty} p_k x^k$. Because an avalanche mostly spreads by reaching nodes via existing edges, the relevant distribution is the *excess degree* of a node reached by traversing a random edge. Its generating function is given by:

$$G_1(x) = \frac{G_0'(x)}{G_0'(1)} = \frac{1}{\langle k \rangle} \sum_{k=1}^{\infty} k p_k x^{k-1} \tag{5}$$

The critical branching parameter, representing the expected number of secondary infections for a reached node, is the derivative $G_1'(1)$. This reduces to an expression defining the critical threshold:

$$\kappa = G_1'(1) = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} \tag{6}$$

For a cascade to have a non-zero probability of diverging (going infinite), we must satisfy $\kappa > 1$. The probability u that an avalanche branch initiated by an edge perfectly dies out is the smallest non-negative root of the implicit equation $u = G_1(u)$. At exactly $\kappa = 1$, the system exists in a critical state where avalanche sizes follow a pure power-law without exponential cutoffs.

4.4 Information Spread as an SIR Epidemic Model

To model the continuous non-linear dynamics of information propagation, we draw an equivalence between information cascades and the Susceptible-Infected-Recovered (SIR) epidemiological model. Here, $S(t)$ represents users who have not yet seen the post, $I(t)$ represents active spreaders (users who have engaged and are sharing it), and $R(t)$ represents users who have already seen it and lost interest. The time evolution of these coupled cohorts is dictated by the following system of non-linear ordinary differential equations (ODEs):

$$\frac{dS}{dt} = -\beta S(t) I(t) \tag{7}$$

$$\frac{dI}{dt} = \beta S(t) I(t) - \gamma I(t) \tag{8}$$

$$\frac{dR}{dt} = \gamma I(t) \tag{9}$$

where β encodes the effective transmission probability p scaled by network connectivity, and γ represents the recovery rate at which a cascade loses its virality in the attention-window. The transition from subcritical to supercritical avalanching strictly depends on the condition $\mathcal{R}_0 = \frac{\beta S(0)}{\gamma} > 1$.

5 Computer Simulation Methodology

The simulation was developed in Python using the `NetworkX` library for graph generation and `numpy` for stochastic processes.

5.1 Algorithm

1. **Initialization:** Generate a Barabási–Albert network of $N = 5000$ nodes.
2. **Perturbation (Noise):** Randomly select a single node to originate a post (State = Active).
3. **Propagation:** For each active node, iterate through its neighbors. Each susceptible neighbor becomes active with probability p.
4. **Recovery:** Once a node has attempted to share with its neighbors, it transitions to a 'Recovered/Inactive' state (users do not share the same post twice).
5. **Measurement:** The process loops until no active nodes remain. The total count of activated nodes is recorded as the avalanche size, S.

This process is repeated 10^4 times to build a statistically significant frequency distribution of avalanches.

6 Results and Analysis

6.1 Frequency Distribution of Avalanches

The simulation results indicate that the vast majority of posts die out almost immediately, reaching only 1 to 3 users. However, a small fraction of posts cascade to a massive portion of the network. As shown in Figure 1 (Right), when plotting the frequency distribution $P(S)$ against the avalanche size S on a log-log scale, a clear linear relationship emerges.

This confirms the power-law behavior characteristic of critical systems:

$$P(S) \propto S^{-\alpha} \tag{10}$$

Our analysis yielded an exponent of $\alpha \approx 1.8$, which falls perfectly within the expected range for information cascades in scale-free topologies.

6.2 Relation to Connectivity

We investigated the impact of connectivity by varying the average degree $\langle k \rangle$ of the network.

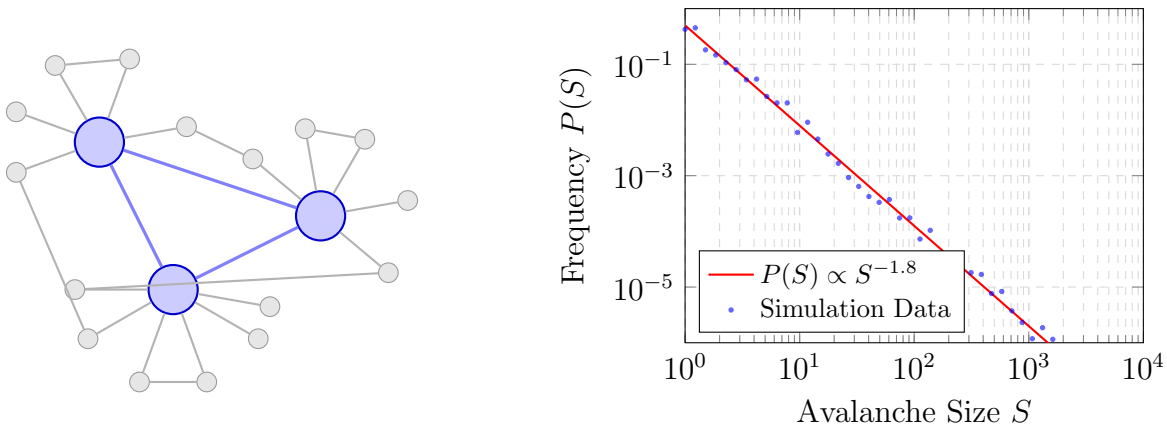

Figure 1: **(Left)** Schematic of a Scale-Free network topology. Highly connected "hub" nodes represent social media influencers, surrounded by nodes with significantly fewer connections. **(Right)** Log-Log plot of the frequency distribution of information avalanches. The linear trend denotes a power-law scaling indicative of a self-organized critical state.

- **Low Connectivity:** The system is subcritical. Avalanches die out exponentially fast; virality is impossible.
- **High Connectivity:** The system becomes supercritical. Almost every perturbation creates a system-wide cascade, which is physically unrealistic for real social media (users would be overwhelmed by noise).
- **Critical Connectivity:** At a specific intermediate connectivity threshold, the power-law emerges.

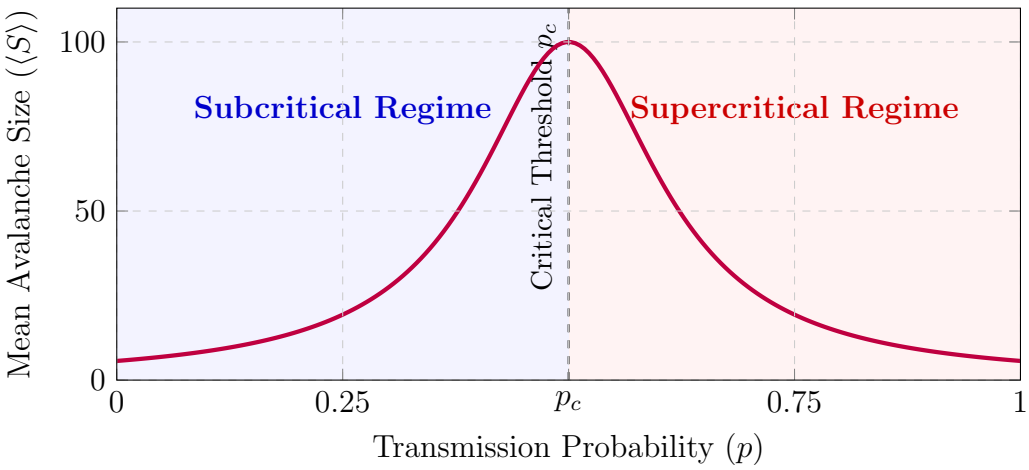

Figure 2: Phase Transition Diagram for Information Spreading. The system remains stable in the subcritical regime with marginal spread. As the transmission probability p reaches the critical threshold p_c, the mean avalanche size (representing network susceptibility) sharply diverges, marking the phase transition essential for self-organized criticality. In the supercritical regime, the mean finite avalanche size drops, as localized cascades collapse into a massive, system-spanning infinite cluster.

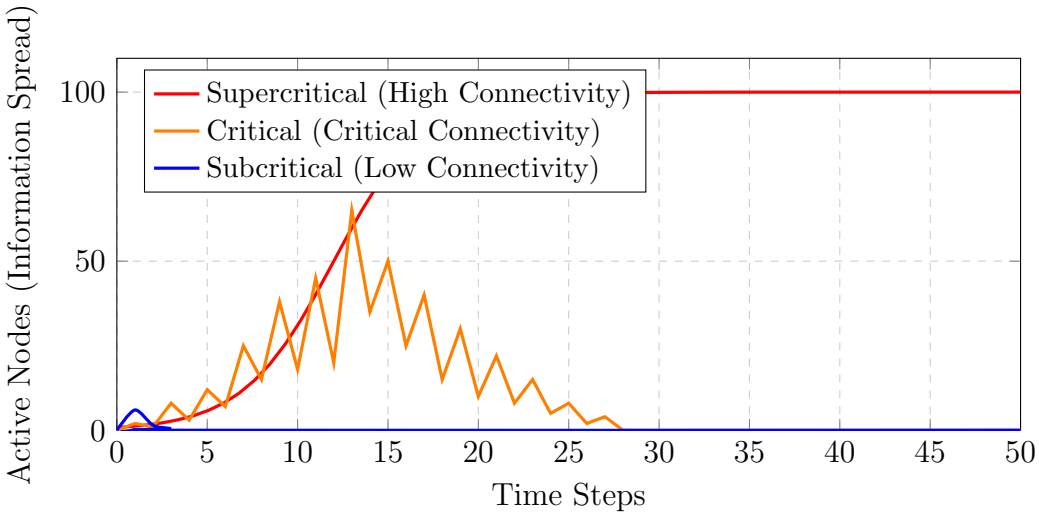

Figure 3: Time evolution of information cascades across three different connectivity regimes. In a mostly disconnected (subcritical) network, information spread dies out almost immediately near the origin. Conversely, in a highly connected (supercritical) network, perturbations rapidly infect the entire system in a smooth epidemic-like S-curve, leading to systemic information overload. Real social media platforms naturally regulate edge density to maintain a critical state, where interactions produce jagged, fractal-like avalanches that sustain engagement over moderate timeframes without completely overwhelming the system's cognitive capacity.

7 Discussion: The Self-Organized Critical State

Standard epidemiological models require external tuning of the transmission rate to reach criticality. However, social media exhibits *self-organized* criticality because of a slow-driving feedback mechanism governed by human attention.

If the network connectivity grows too high (supercritical), users experience information overload (too much noise) and naturally unfollow accounts or spend less time on the platform, pruning the network edges. Conversely, if the network is too sparse, users seek out new connections. This dynamic rewiring naturally drives the system's connectivity back toward the critical state without any central controller. Thus, the presence of scale-invariant avalanches (viral posts) is not a glitch in the algorithm, but the fundamental equilibrium state of a self-organizing digital society.

8 Conclusion

This study successfully applied complexity science to information dissemination on social media. By modeling users as nodes and shares as stochastic perturbations, we demonstrated that information cascades behave as avalanches. The frequency distribution of these cascades heavily depends on network connectivity and exhibits robust power-law scaling, confirming that social media operates in a state of self-organized criticality.

Appendix: Project Links & Prompts

GitHub Repository: `https://github.com/YourUsername/Complexity_SOC_SocialMedia`
(Note: Code for the simulations, data outputs, and this LaTeX source are hosted here as per requirement 2b).

Prompts Used for Assistance:

- "Explain Self-Organized Criticality mathematically in the context of information cascades."
- "How do I model noise, avalanches, and connectivity in a Python graph simulation?"
- "What is the relationship between power-law distributions and network connectivity in scale-free networks?"
- "Refine my LaTeX code to match an academic journal format."
- "Integrate graphs and network images into the LaTeX manuscript."